DR6h: A Target Without a Domain-General Proxy Narrows the Margin

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair — DR6h **Status:** Overall **NO_GO** on the "wall bites" gate suite. But the margin degraded significantly: realisations spanned 5–10 (stdev 2.35, an entire 2 sigma wider than any prior DR6 variant), and two realisations (RH3 append_log and RH4 rename_after_write) scored 5–6, approaching placebo territory. The wall is *starting to bite* — DR5*'s domain-general-proxy hypothesis is directionally right but the LLM's residual semantic reasoning still avoids direct overlap. **Date:** 2026-07-27

Abstract

DR6g refined DR5 to DR5*: the wall bites when the verifier has no signal — semantic reasoning about D or a domain-general proxy — that separates realisations from non-realisations. DR6g showed the implicit-vs-explicit proxy escapes the wall for naive-UTC because that D literally IS about implicit assumptions. DR6h tests DR5*'s prediction that targets without such a proxy hit the wall even for LLM verifiers with D specified.

Target D_h: "*This code assumes exclusive access to files it reads or writes — no other process, thread, or async task will read, write, truncate, or delete those files concurrently.*" No clean implicit-vs-explicit proxy: both correctly-locked and dangerously-unlocked file access can be terse or verbose.

Result: partial wall, not direct wall. Overlap gap +2 (down from DR6e's +7 on naive-UTC). Realisation stdev 2.35 (up from DR6e's 0.98). Two realisations (RH3 append_log scored 6, RH4 rename_after_write scored 5) sit near the placebo distribution's top (PH4 read_only_open scored 3). Claude still discriminated well enough to avoid direct realisation-placebo overlap, but the margin is narrower than every prior DR6 variant.

*DR5, refined again (DR5): LLM semantic access provides a signal whose *strength* depends on (i) how deep D's semantic definition is, (ii) whether the target has a domain-general surface correlate, and (iii) how much of the LLM's training corpus covers reasoning about D in code contexts. On D_h, the LLM has (i) partial (the definition requires reasoning about concurrent access, which is a well-covered domain in the LLM's training) but not (ii) (no clean surface correlate), so discrimination degrades — the wall's *shadow* appears without the wall itself.

DCR1f's T1 (absolute simultaneity in pre-1905 physics) fails on all three properties for a 1904 verifier: (i) T1's presupposition-structure was invisible to physicists at the time; (ii) no domain-general correlate distinguishes T1 from T2 or from non-realisations; (iii) the LLM's training corpus post-dates T1's deletion by 121 years. Every condition of DR5**'s escape route fails. Wall bites hard.

The empirical picture from DR6d/e/f/g/h is that the wall's severity is *graded*, not binary, and the grading is captured by DR5**'s three factors. What DR5 called "the wall" is the extreme end of a spectrum; what DR6 escapes are the softer regions of the same spectrum.

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1. Setup

10 new snippets in snippets_h.py:

Realisations (5) of D_h:

- RH1: read-modify-write with no locking
- RH2: seek+write pattern, single handle

- RH3: append-to-log without lock
- RH4: write-then-rename atomic replace, no coordination on the read side
- RH5: JSON load-modify-dump

Placebos (5):

- PH1: fcntl.flock exclusive lock
- PH2: filelock library, explicit
- PH3: no file access at all (in-memory)
- PH4: read-only open, side-effect free
- PH5: hand-rolled pidlock with retry

Three sandboxed Claude subagents. D_h specified verbatim in prompt.

2. Results

Per-snippet consensus scores:

snippet	kind	per verifier	consensus
RH1_readmodifywrite	realisation	(from JSON)	**10**
RH2_seek_write	realisation		**9**
RH3_append_log	realisation		**6**
RH4_rename_after_write	realisation		**5**
RH5_json_load_modify_dump	realisation		**10**
PH1_fcntl_lock	placebo		**1**
PH2_filelock_library	placebo		**0**
PH3_no_file_access	placebo		**0**
PH4_read_only_open	placebo		**3**
PH5_lockfile_context	placebo		**1**

Aggregate:

- Realisations: [10, 9, 6, 5, 10] — median 9, stdev **2.35**, min 5, max 10.
- Placebos: [1, 0, 0, 3, 1] — median 1, max 3.
- Overlap gap: 5 − 3 = **+2**.

3. Gate decisions

gate		
W1 completeness	GO	
W2 realisation median ≥ 6	GO	9
W3 realisation stdev ≥ 1.5	**GO**	2.35 (wall indicator!)
W4 placebo median ≤ 3	GO	1
W5 placebo trigger ≥ 5	NO_GO	max 3
W6 direct overlap	NO_GO	gap +2

Overall NO_GO on the strict "wall bites" gate. But W3 GO is significant — realisation variability is 2× to 3× wider than every prior DR6 variant. The wall is starting to bite in the variance dimension without producing direct overlap yet.

Reading: clean_discrimination_despite_no_obvious_proxy. Modulated: LLM semantic reasoning succeeded in absolute terms but at reduced margin, exactly as DR5*'s directional prediction anticipates.

4. Comparison with prior DR6 variants

paper	target D	prompt	overlap gap	realisation stdev
DR6	naive-UTC	D specified	7	0.98
DR6e	naive-UTC (extended)	D specified	7	0.98
DR6f	naive-UTC	D-adjacent	6	0.82
DR6g	naive-UTC	fully blind	6	0.63
DR6h	**exclusive file access**	**D specified**	**2**	**2.35**

The stdev jump from ≤1 to 2.35 is the strongest signal. DR6h's realisations span the widest range of scores because the LLM is reasoning about each snippet's semantic content and reaching different conclusions: RH1/RH2/RH5 are "obvious" (multi-step file operations, clear concurrency risk) while RH3/RH4 are subtler (append is atomic on most filesystems for small writes; write-then-rename is a *pattern for* concurrency-safety though the read side is still unlocked).

Prior DR6 targets could be scored via a domain-general proxy that gave uniform-high scores to all realisations. DR6h's target has no such proxy, so realisations get scored on their individual semantic content — which varies.

5. What DR6h establishes

fully bite even without a domain-general proxy — LLM semantic reasoning provides substantial signal. But margin narrows and variability increases exactly where DR5* predicts.

domain-general correlate, LLM training coverage) each modulate the margin. Targets fail all three (DCR1f T1) bite hard; targets satisfy all three (DR6e naive-UTC with D specified) escape cleanly; targets in between produce intermediate outcomes.

DR6 variants had stdev ≤ 1; DR6h jumped to 2.35. High stdev signals the wall is starting to bite even before direct overlap appears.

- **DR5* is directionally right but too strong.** The wall does not
- **The wall is graded.** DR5**'s three factors (semantic depth of D,
- **The single most predictive metric is realisation stdev.** All prior

6. Refined-refined claim (DR5**)

The wall's severity is a graded function of three factors, all verifier-relative:

commitment in a form the verifier can reason about.

feature (implicit-vs-explicit, structured-vs-unstructured, etc.) that discriminates realisations from non-realisations without requiring D-specific reasoning.

covers reasoning about D-like commitments in the target domain.

1. **Semantic depth of D** — how well-defined is the target
2. **Domain-general correlate** — does the target admit a surface
3. **LLM training coverage** — how much of the LLM's training corpus

The wall is severe when all three fail. The wall is absent when all three succeed. Intermediate cases (DR6f/g/h) produce intermediate outcomes: some discrimination, but at reduced margin and with high variability across realisations of the same target.

7. What DR6h does not license

Proposition-ranking N still cannot distinguish D from r_i under DR5's antecedents; DR6h just shows the LLM's escape via condition (b) is graded, not binary.

the observation. A preregistered stdev threshold might catch the wall's degrees more cleanly in a future DR8 gradient study.

- **DR5's original structural theorem is refuted.** It is not.

- **The stdev metric is universal.** DR6h introduced it post-hoc from

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Appendix: reproduction

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uv run --no-sync python -m experiments.dr6_code_correctness_corollary.run_dr6h
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Deterministic scoring over three verifier JSON files, six-gate output in results/dr6h_verdict.json.